Measurement of Growth and Development

Part 1 : Physical Growth

Third Stage

Dr. Moustafa Alshammari

Pediatric lab

Anthropometric measurements

Anthropometric measurements are noninvasive quantitative measurements of the body. Essential parameter of **nutritional** status, is the measurement of height or length, weight, head circumference, chest circumference proportions, skinfold thickness, and arm circumference in young children. Height and head circumference reflect **past nutrition**, whereas weight, skinfold thickness, and arm circumference reflect **present nutritional** status, especially of protein and fat reserves.

(continued)

Growth Measurements

Measurement of **physical growth** in children is a key element in evaluating their health status. Physical growth parameters include **weight**, **height** (**length**), **skinfold thickness**, **arm circumference**, and **head circumference**. Values for these growth parameters are plotted on percentile charts, and the child's measurements in percentiles are compared with those of the general population.

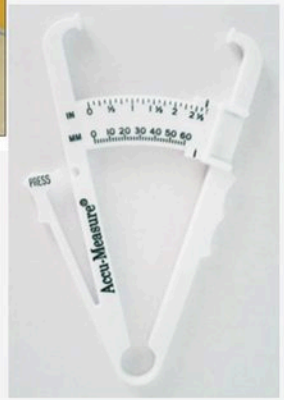
(continued)

Equipment





Fig. 2. Height measurement. (Image shown with consent.)





* Weight scale
* Stadiometer
* Skinfold calipers
* Non-stretchable tape measure
* Infantometer to measure the recumbent length

Technique

* Growth Charts
* Length
* Height
* Weight
* Skinfold Thickness and Arm Circumference
* Head Circumference
* Chest circumference:
* Body Mass Index

Growth Charts

* Growth charts use a series of percentile curves to demonstrate the distribution of body measurements in children.

Growth Charts

* Children whose growth may be questionable include:

1. Children whose height and weight percentiles are widely disparate (e.g., height in the 10th percentile and weight in the 90thercentile, especially with above-average skinfold thickness)

2. Children who fail to follow the expected growth velocity in height and weight, especially during the rapid growth periods of infancy and adolescence

Growth Charts

3. Children who show a sudden increase (except during puberty), decrease, or no change in a previously steady growth pattern

* Because growth is a continuous but uneven process, the most reliable evaluation lies in comparing growth measurements over time. It is important to remember that normal growth patterns vary among children the same age

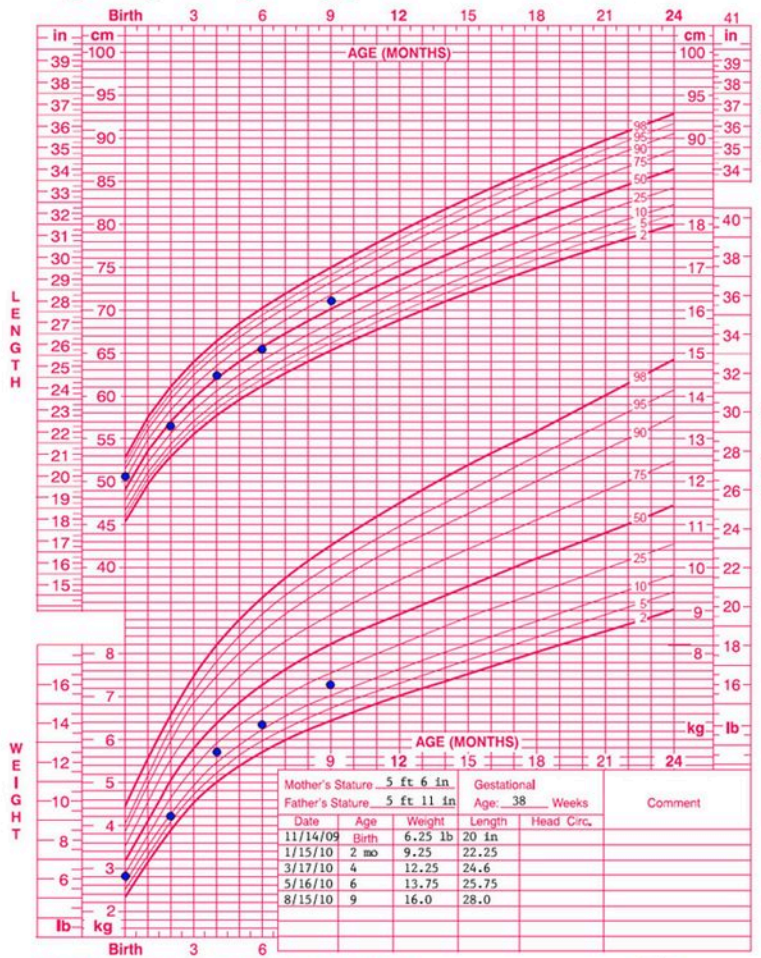


* These children of identical age (8 years) are markedly different in size. Child on left, of Asian descent, is at 5th percentile for height and weight. Child on right is above 95th percentile for height and weight. However, both children demonstrate normal growth patterns.

Birth to 24 months: Girls
Length-for-age and Weight-for-age percentiles

NAME Maya

RECORD #



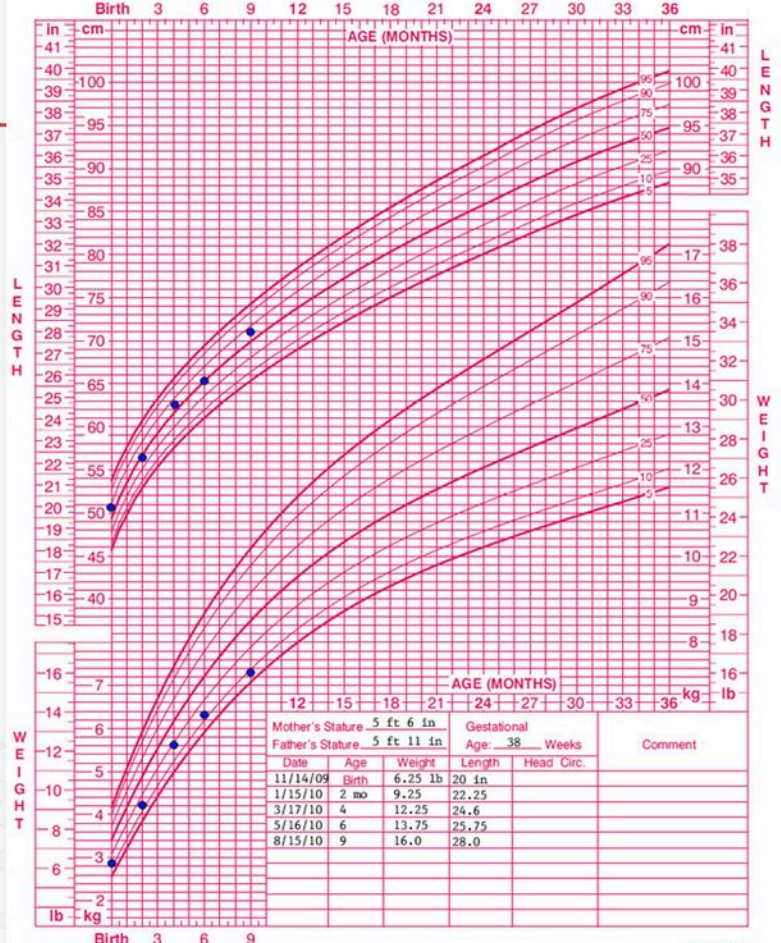
Mother's Stature 5 ft 6 in			Gestational		
Father's Stature 5 ft 11 in			Age: 38 Weeks		Comment
Date	Age	Weight	Length	Head Circ.	
11/14/09	Birth	6.25 lb	20 in		
1/15/10	2 mo	9.25	22.25		
3/17/10	4	12.25	24.6		
5/16/10	6	13.75	25.75		
8/15/10	9	16.0	28.0		

Published by the Centers for Disease Control and Prevention, November 1, 2009
SOURCE: WHO Child Growth Standards (http://www.who.int/childgrowth/en)

Birth to 36 months: Girls
Length-for-age and Weight-for-age percentiles

NAME Maya

RECORD #



Mother's Stature 5 ft 6 in			Gestational		
Father's Stature 5 ft 11 in			Age: 38 Weeks		Comment
Date	Age	Weight	Length	Head Circ.	
11/14/09	Birth	6.25 lb	20 in		
1/15/10	2 mo	9.25	22.25		
3/17/10	4	12.25	24.6		
5/16/10	6	13.75	25.75		
8/15/10	9	16.0	28.0		

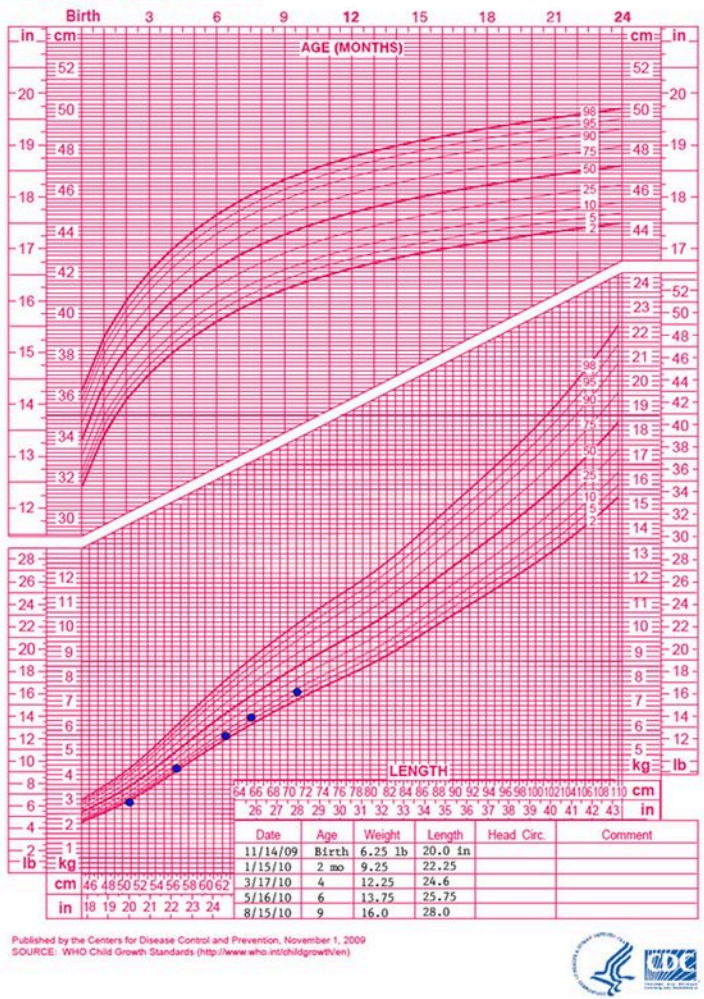
Published May 30, 2000 (modified 4/20/01).
SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
http://www.cdc.gov/growthcharts

SAFER · HEALTHIER · PEOPLE™

Birth to 24 months: Girls
Head circumference-for-age and Weight-for-length percentiles

NAME Maya

RECORD # _____



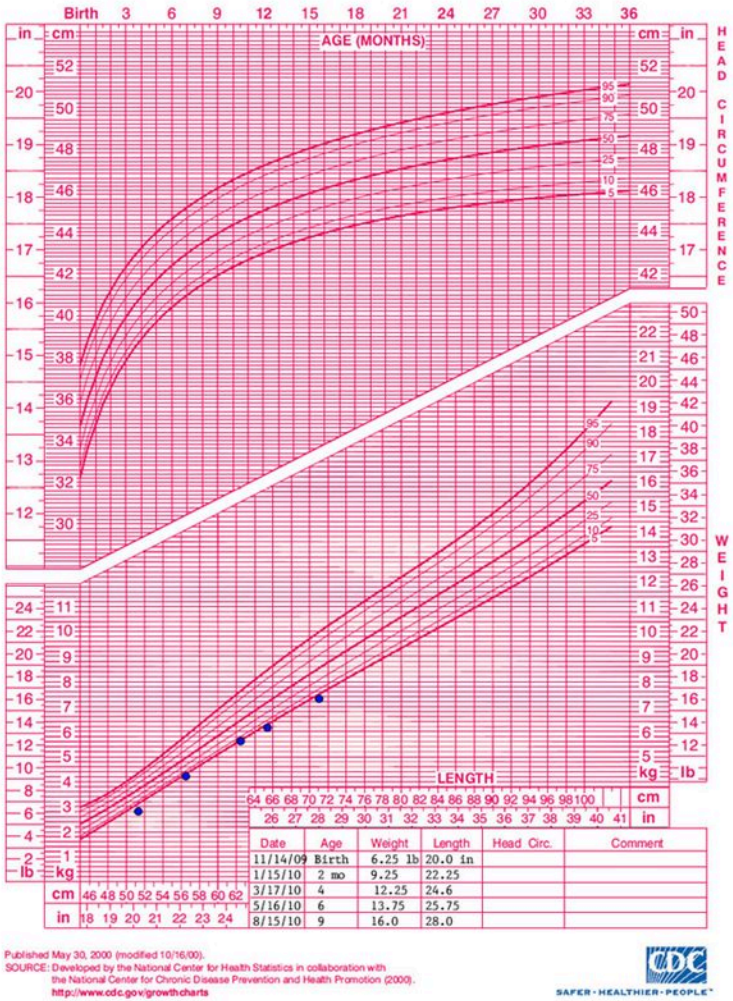
Date	Age	Weight	Length	Head Circ.	Comment
11/14/09	Birth	6.25 lb	20.0 in		
1/15/10	2 mo	9.25	22.25		
3/17/10	4	12.25	24.6		
5/16/10	6	13.75	25.75		
8/15/10	9	16.0	28.0		

Published by the Centers for Disease Control and Prevention, November 1, 2009
SOURCE: WHO Child Growth Standards (http://www.who.int/childgrowth/en)

Birth to 36 months: Girls
Head circumference-for-age and Weight-for-length percentiles

NAME Maya

RECORD # _____



Date	Age	Weight	Length	Head Circ.	Comment
11/14/09	Birth	6.25 lb	20.0 in		
1/15/10	2 mo	9.25	22.25		
3/17/10	4	12.25	24.6		
5/16/10	6	13.75	25.75		
8/15/10	9	16.0	28.0		

Published May 30, 2000 (modified 10/16/00).
SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
http://www.cdc.gov/growthcharts

SAFER · HEALTHIER · PEOPLE™

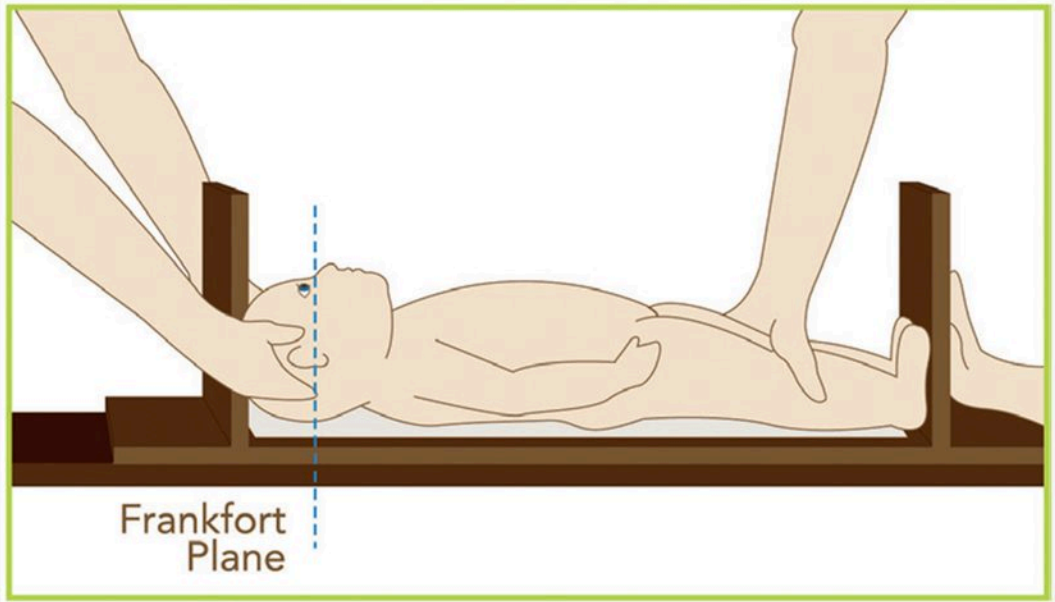
Length

- The term length refers to measurements taken when children are supine
- Length in infants are measured by lying in supine position on tray or board.
- Hold the infant's head in midline and extend the hips and knees flat on a paper-covered table, then Points are marked at the top of the head and the heels of the feet
- Birth – 6 months: Monthly gain—2.5 cm (1 inch)
- 6-12 month: Monthly gain—1. 5 cm (0.5 inch)

Measurement of Length

Newborn =48 to 53 cm (18.9 to 20.9 inches).

Infant =75 cm (30 inches).



Height

* Height (or stature) refers to the measurement taken when a child is standing upright.

* Measure height by having the child, with shoes removed, stand as tall and straight as possible, with the head in midline and the line of vision parallel to the ceiling and floor.

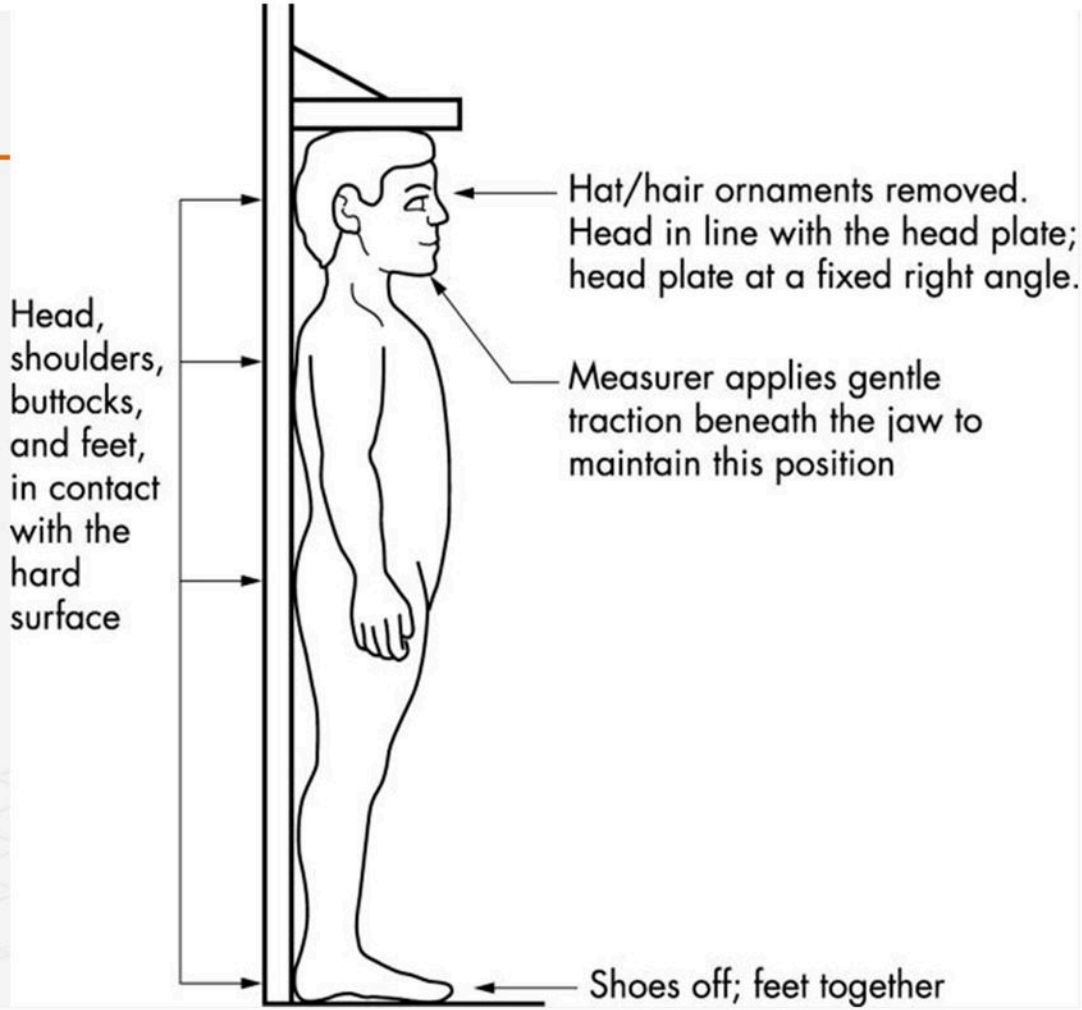
Height

- Be certain the child's back is to the wall or other vertical flat surface, with the heels, buttocks, and back of the shoulders touching the wall.
- Check for and correct bending of the knees, slumping of the shoulders, or raising of the heels.
- Normally, height is less if measured in the afternoon than in the morning. The time of day should be recorded when measurements are . For children in whom there are concerns about growth, serial measurements should be taken at the same time of day to establish an accurate growth velocity.

Height

* For the most accurate measurement, use a wall-mounted unit (stadiometer). The movable measuring rod of platform scales is accurate only if it remains parallel to the floor and rests securely on the topmost part of the head.
* To improvise a flat surface for measuring length, attach a paper or metal tape or yardstick to the wall, position the child adjacent to the tape, and place a three-dimensional object, such as a thick book or box, on top of the head.
* Rest the side of the object firmly against the wall to form a right angle. Measure length or stature to the nearest 1 mm or 1/8 inch.

Measurement of height



Weight

* At birth, weight 2700 to 4000 kg (6–9 pounds).
* First, balance the scale.
* Remove the infant's diaper and clothing.
* Depending on age, put the child is either supine or sitting.

Weight

- When precise measurements are necessary, two nurses should take the weight independently; if there is a discrepancy, take a third reading.
- Take measurements in a comfortably warm room.
- When birth–to–36 months growth charts are used, children should be weighed nude.
- Older children are usually weighed while wearing their underpants or a light gown.

Weight

* Protect the infant from an accidental fall by placing a hand over the infant without direct contact.
* At birth, weight is more variable than height and is, to a greater extent, a reflection of the intrauterine environment.

Weight

* Normally, neonates lose about 10% of their birth weight by 3 to 4 days of age because of loss of extracellular fluid and meconium, as well as limited food intake, especially in breastfed infants.

* In general, the birth weight doubles by 4 to 7 months of age, triples by the end of the first year. Quadruples by the age of 2 to 2.5 years.

Weight



Infant on scale.



Toddler on scale. Note presence of nurse to prevent falls.

Weight progression (1-12 month)

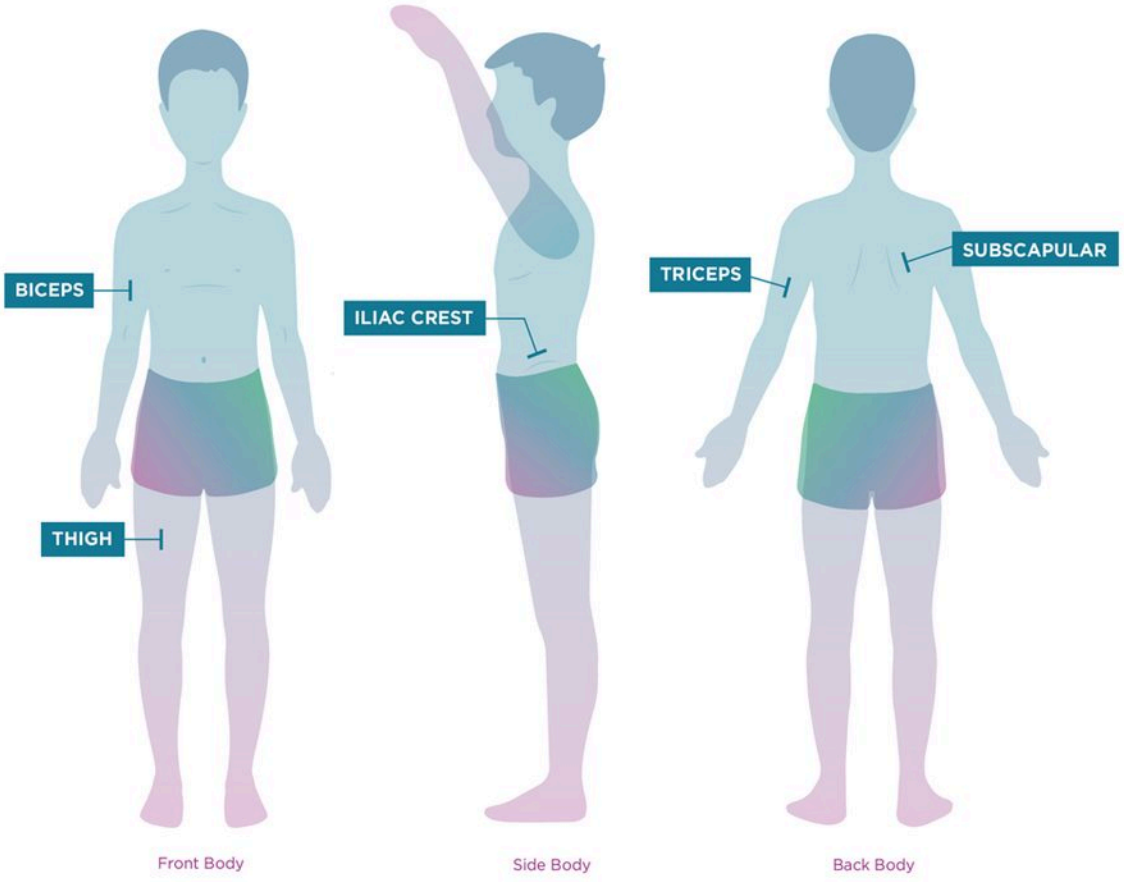
* First 4 month = 750 gm /month
* 2nd 4 months = 500 gm /month
* 3rd 4 months = 250 gm/month
* Progression in year: Weight (kg/years) = (age + 4) ×2.

Ex: 2-year-old will weigh: (2 + 4) ×2 = 12kg.

Skinfold Thickness

* Measures of relative weight and stature cannot distinguish between adipose (fat) tissue and muscle. One convenient measure of body fat is skinfold thickness,

* Measure skinfold thickness with special calipers, such as the Lange calipers. The most common sites for measuring skinfold thickness are the triceps (most practical for routine clinical use), subscapula, suprailiac, abdomen, and upper thigh.


BICEPS
THIGH
ILIAC CREST
TRICEPS
SUBSCAPULAR
Front Body
Side Body
Back Body

Skinfold Thickness

* For greatest reliability, follow the exact procedure for measurement, and record the average of at least two measurements of one site.

Head Circumference

- Head circumference is a critical indicator of brain growth in infants and toddlers.
- Full-term infants, average head circumference is between 33 and 35.5 cm (13 and 14 inches).
- Measure head circumference in children up to 36 months of age
- The rate of increase in head circumference slows somewhat by the end of infancy, and head circumference is usually equal to chest circumference by 1 to 2 years of age.

Head Circumference

* The usual total increase in head circumference during the second year is 2.5 cm (1 inch). Then the rate of increase slows until at age 5 years, the increase is less than 1.25 cm (0.5 inch) per year.

* The head's largest circumference is measured by placing the tape over the lower forehead, above the pinna of the ears, and over the occipital prominence.

Head Circumference

* Head circumference is usually about 2 to 3 cm (≈1 inch) greater than chest circumference.

* Because of the molding of the head during delivery, these measurements may initially appear equal. **microcephaly** or premature closure of the sutures (**craniosynostosis**) is a possibility. The head is more than 4 cm (1.6 inches) larger than the chest and this remains constant or increases over several days, then hydrocephalus must be considered.

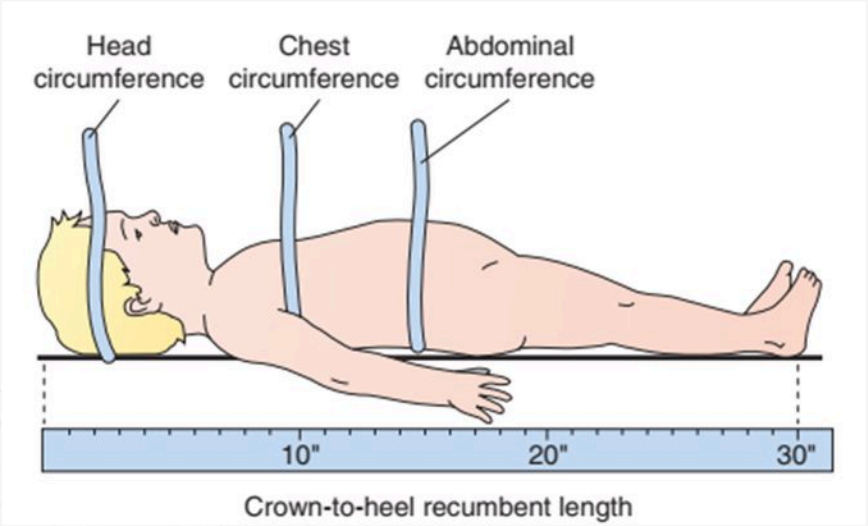
Head Circumference

* During the first 6 months, head circumference increases approximately 1.5 cm (0.6 inch) per month, but the rate of growth declines to only 0.5 cm (0.2 inch) monthly during the second 6 months.

* Closure of the cranial sutures occurs, with the posterior fontanel fusing by 6 to 8 weeks of age and the anterior fontanel closing by 12 to 18 months of age (average, 14 months).

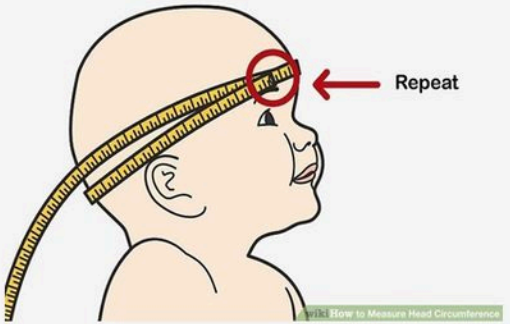
Head Circumference

Measurement of head, chest, and abdominal circumference and crown-to-heel (recumbent) length.



Head Circumference

Measurement of head,



* H.C (Birth – 6 months): Monthly gain—1.5 cm
* H.C (6-12 month): Monthly gain—0. 25 cm

Body Mass Index

* BMI is also a part of assessing growth and can be tracked over time.
* BMI is the number calculated from the child's weight and height that correlates to the percentage of body fat. The higher the child's BMI, the higher the percentage of body fat.

Parameters

* Underweight: BMI of less than 18.5
* Normal weight: BMI of 18.5 to 24.9
* Overweight: BMI of 25 to 29.9
* Obesity: BMI of 30 or greater

Body Mass Index Formula

(BMI [weight in kilograms divided by height in meters squared])

BMI= Weight in kilograms÷ (Height in meters)2

Example Calculation:**7-year-old** child with the following measurements:

Weight: **25 kg**

Height: **1.2 meters** (120 cm)

The BMI calculation would be:

$$\text{BMI}=\frac{25}{(1.2)^2}=\frac{70}{1.44}\approx 17.36$$

Thank you